

```
close all;
clear;
clc;
% Read image
image=imread("dog_backpack.png");
% Convert image into grayscale
gray=rgb2gray(image);
% Get size of both image
size(image)
size(gray)
% Show both image side by side
figure;
subplot(1,2,1);
imshow(image);
subplot(1,2,2);
imshow(gray);
```

ans =

1401 934 3

ans =

1401 934



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% Extract height and width of image and print it
[m,n,~]=size(gray);
fprintf("dimension is %d * %d \n",m,n);
% Get datatype of image
type=class(gray);
fprintf('Datatype of the image is %s\n',type);
% Find total pixel in image;;
total=n*m;
fprintf("total pixel of the image is %d \n",total);
% Find dynamic range of the image
minvalue=min(gray(:));
maxvalue=max(gray(:));
fprintf("dynamic range of the image is %d to %d \n",minvalue, maxvalue);

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dimension is 1401 * 934
Datatype of the image is uint8
total pixel of the image is 1308534
dynamic range of the image is 9 to 255

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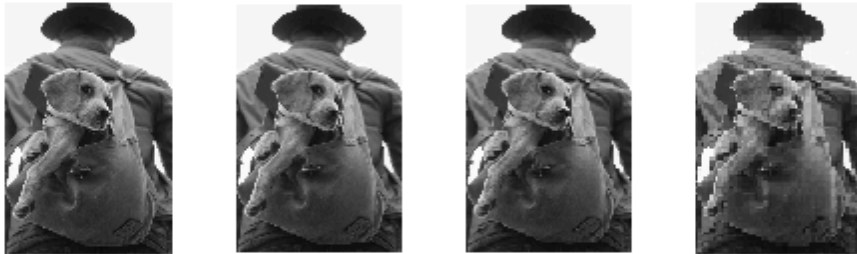
% Print dimension of original image %d
fprintf("Dimension of the original image is %d %d \n",size(gray,1),size(gray,2));
% Create how many factors i need to subsample image by 4 , 8 and 16
ka=4;
kb=8;
kc=16;
% Downsample the image
ka_image=imresize(gray,1/ka,'nearest');
kb_image=imresize(gray,1/kb,"nearest");
kc_image=imresize(gray,1/kc,"nearest");
% Print out the dimension after downsampling
fprintf("dimension of image after downsampling by 4 factor % d %d \n",size(ka_image,1),size(ka_image,2));
fprintf("dimension of image after downsampling by 8 factor % d %d \n",size(kb_image,1),size(kb_image,2));
fprintf("dimension of image after downsampling by 16 factor % d %d \n",size(kc_image,1),size(kc_image,2));
% Upsample image to see the changes
ka_resize=imresize(ka_image,[m n],"nearest");
kb_resize=imresize(kb_image,[m n],"nearest");
kc_resize=imresize(kc_image,[m n],"nearest");
figure;
subplot(1,4,1);
imshow(gray);
subplot(1,4,2);
imshow(ka_resize);
subplot(1,4,3);
imshow(kb_resize);
subplot(1,4,4);
imshow(kc_resize);

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Dimension of the original image is 1401 934
dimension of image after downsampling by 4 factor  351 234
dimension of image after downsampling by 8 factor  176 117
dimension of image after downsampling by 16 factor  88 59

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% Create function which convert image into what level that we want
quant= @(i,k) uint8(floor(double(i)/(256/k))*(256/k));
% Call that function with level 128 32 8
level_128=quant(gray,128);
level_32=quant(gray,32);
level_3=quant(gray,3);

% Create image with level 2
level_2=uint8(gray>=128)*255;

figure;
subplot(1,4,1);
imshow(gray);
subplot(1,4,2);
imshow(level_128);
subplot(1,4,3);
imshow(level_3);
subplot(1,4,4);
imshow(level_2);
```

